HW9

P1. (a) $$X(jw) = \int_0^1 (1-t^2)\, e^{-jwt}\, dt$$

$$= \int_0^1 e^{-jwt}\, dt - \int_0^1 t^2 e^{-jwt}\, dt$$

$$= \frac{e^{-jwt}}{-jw}\Big|_0^1 - \left[t^2 \frac{e^{-jwt}}{-jw} - 2t\frac{e^{-jwt}}{-w^2} + \frac{2e^{-jwt}}{jw^2}\right]_0^1$$

$$= \frac{e^{-jw}}{-jw} + \frac{1}{jw} - \left[\frac{e^{-jw}}{-jw} + \frac{2e^{-jw}}{w^2} + \frac{2e^{-jw}}{jw^3}\right] + \frac{2}{jw^3}$$

$$= \frac{1}{jw} - \frac{2e^{-jw}}{w^2} - \frac{2e^{-jw}}{jw^3} + \frac{2}{jw^3}$$

when $w=0$ $\quad X(0) = \int_0^1 (1-t^2)\,dt = \frac{2}{3}$

$$\therefore\ X(jw) = \begin{cases} \frac{2}{3} & w=0 \\ \frac{1}{jw} - \frac{2e^{-jw}}{w^2} - \frac{2e^{-jw}}{jw^3} + \frac{2}{jw^3} & w\neq 0 \end{cases}$$

(b) $$g(t) = e^{-3|t|} = \begin{cases} e^{-3t}, & t>0 \\ e^{3t}, & t<0 \end{cases}$$

$$g(t) \xrightarrow{F} G(jw) = \int_0^\infty e^{-(3+jw)t}\, dt + \int_{-\infty}^0 e^{(3-jw)t}\, dt$$

$$G(jw) = \frac{1}{3+jw} + \frac{1}{3-jw} = \frac{6}{9+w^2}$$

$$\sin 2t = \frac{e^{j2t} - e^{-j2t}}{2j}$$

$$x(t) = \frac{g(t)e^{j2t} - g(t)e^{-j2t}}{2j}$$

$$X(jw) = \frac{1}{2j}\left[G(j(w-2)) - G(j(w+2))\right]$$

$$= \frac{1}{2j}\left[\frac{3}{9+(w-2)^2} - \frac{3}{9+(w+2)^2}\right]$$

(c) $$X(jw) = \int_0^\infty t\, e^{-(a+jw)t}\, dt$$

$$= t\,\frac{e^{-(a+jw)t}}{-(a+jw)} - \frac{e^{-(a+jw)t}}{(a+jw)^2}\Big|_0^\infty$$

$$= \frac{1}{(a+jw)^2}$$

P2. (a) $$y(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} Y(jw)\, e^{jwt}\, dt$$

$$F(x(t)y(t)) = \int_{-\infty}^{\infty} \frac{1}{2\pi} x(t) \int_{-\infty}^{\infty} Y(jw) e^{jwt} e^{-jvt}\, dt$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} Y(jw) \int_{-\infty}^{\infty} x(t) e^{j(w-v)t}\, dt$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} Y(jw)\, X(w-v)$$

$$= \frac{1}{2\pi} X(jw) * Y(jw)$$

(b) $$x(t) * y(t) = \int_{-\infty}^{\infty} x(\tau)\, y(\tau - t)\, d\tau$$

$$F(x(t)*y(t)) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x(\tau)\, y(\tau-t)\, e^{-jwt}\, d\tau\, dt$$

$$= \int_{-\infty}^{\infty} x(\tau) e^{-jw\tau}\, d\tau \int_{-\infty}^{\infty} y(\tau-t)\, e^{-jw(t-\tau)}\, dt$$

$$= \int_{-\infty}^{\infty} x(\tau) e^{-jw\tau}\, d\tau \int_{-\infty}^{\infty} y(\tau-t)\, e^{-jw(t-\tau)}\, d(t-\tau)$$

$$= X(jw)\, Y(jw)$$

Heyang Zhang bracket_x@163.com